

Teaching Statement

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Three Universities, One Philosophy

I have taught since 2014 at three very different universities: large undergraduate cohorts at Helwan University in Cairo (2014–2018), graduate and undergraduate courses at the National University of Singapore (2019–2023), and the Static Program Analysis course at NYU Abu Dhabi (2025–2026). Each place taught me something different about students, but all three left me with the same conviction. **Students learn security and systems by doing.** Until someone has exploited a vulnerable program, defenses are an abstraction to them, and until they have patched one and watched the patch hold, the skill is not really theirs.

How I Teach

Break it first, then build the fix. Every concept in my courses comes with a lab where students attack, patch, and re-test a real system. I use capture-the-flag exercises, deliberately vulnerable applications such as OWASP Juice Shop, and testing tools such as OWASP ZAP. In blockchain courses, students reproduce real exploit transactions from the attack corpus I curated for my own research [1], then write and verify the patch that would have stopped the attack. My newest work turns into the lab I most want to teach. Students take a contract behind one of the largest DeFi losses on record, retrofit it with the Authorization Gateway from my framework [2], and run both the attacker and a legitimate user against it under a real EVM. Within one session they watch the attack revert at the guard while the honest call goes through. That lands in a way no slide does, because the defense is visibly changing what the machine will do.

Teach from research, recruit into research. My courses import live material from my publications: real vulnerable contracts from my USENIX Security work [3], real analyzer failures from my MSR study [4], and hand-verified exploit benchmarks from my detection work [5]. I wrote my research agenda with students in mind: my solo paper on smart contract security beyond detection breaks four open directions into scoped capstone and final-year projects [6], so a student choosing a graduation topic picks from live research questions rather than from exercises. Traffic goes the other way too, and I already have it running. Three undergraduate research assistants work with me at NYUAD at the moment, and each year I sit as secondary advisor on three or four capstone and graduation projects. At NUS I mentored project teams in formal verification and blockchain security. A student I worked with is a co-author on my MSR 2026 study [4], which is what a strong course project should be able to turn into.

Reach every student. Security classes mix students of very different preparation. I structure labs in graduated tiers with explicit rubrics: every student reaches the core outcomes, and stronger students continue into open-ended challenges. I schedule early, low-stakes assessments so I find struggling students in week three, not at the final. As a native Arabic speaker who teaches in English, I support Arabic-speaking students directly, which matters most in the first year, when the language and the material are both new at once.

What I Have Taught

- **NYU Abu Dhabi** (Spring 2025, Spring 2026): CS-UH 3260 Static Program Analysis. I prepared lab materials and assignments, assisted lectures and grading, and supervised course projects on real codebases.
- **National University of Singapore** (2019–2023): Teaching Assistant for Compiler Design, Programming Languages, Formal Verification, and Blockchain Security. I lectured recitations, designed assignments and exams, and mentored project teams at graduate and undergraduate levels.
- **Helwan University** (2014–2018): Teaching Assistant across the undergraduate curriculum, from first programming courses in C, C++, Java, and Python through data structures, compilers, and software engineering, and later machine learning and blockchain security. I ran labs, tutored, graded, and supervised graduation projects.

What I Am Ready to Teach

At undergraduate level I can take the security core, from an introductory course through network and system security, cryptography, secure development, ethical hacking, digital forensics, malware analysis, and incident response, as well as cloud and IoT security. I can equally cover the computing foundations I taught for years before specialising: programming, data structures and algorithms, compilers, and software engineering. My CV lists these individually.

At graduate level I would teach program analysis, formal verification, and machine learning for security, and I would like to build two new courses. The first is Blockchain and Smart Contract Security, taught from my own systems and datasets. The second, Cyber Risk, Compliance, and Virtual-Asset Regulation, connects the security work to the regulatory frameworks now taking shape across the Gulf [6], and I am not aware of many places teaching it.

A load of 3–4 modules per semester across both levels is what I expect, cyber forensics included, and I am ready to supervise M.Sc. and Ph.D. dissertations in cybersecurity, blockchain, and applied machine learning. I would also expect to take part in program-level curriculum review.

Assessment and Accreditation

I design courses backward from measurable learning outcomes, following **ABET** criteria and the **ACM/IEEE cybersecurity curricular guidelines (CSEC)**. Each semester I map course outcomes to program outcomes, collect per-outcome attainment data, document weaknesses, and feed the changes into the next offering. I also align lab content with the practical skills behind industry certifications: CEH and OSCP-style offensive competencies, SANS/GIAC incident handling and penetration testing, and the governance vocabulary of CISSP and CISM. Graduates then leave with academic depth and with skills the market recognizes, which is the capacity-building UAE Centennial 2071 asks universities to deliver.

Closing

What I want from a graduate is concrete: that they can find a flaw, show why it matters, fix it, and defend the fix under questioning. I would be glad to build that program at United Arab Emirates University.

References

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- [3] Tamer Abdelaziz and Aquinas Hobor. Smart learning to find dumb contracts. In *32nd USENIX Security Symposium*, pages 1775–1792, 2023.
- [4] Tamer Abdelaziz, Salma Alsaghir, and Karim Ali. Where do smart contract security analyzers fall short? In *23rd Int. Conf. on Mining Software Repositories (MSR)*, 2026.
- [5] Tamer Abdelaziz and Karim Ali. TxSentinel: Protocol-agnostic exploit detection on EVM blockchains. Under review, 2026.
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